

Q. Using Pumping Lemma prove that the language $A = \{a^n b^n \mid n \geq 0\}$ is not regular.

→ step 1: Assume that A is regular.

step 2: Pumping length = P

$$S = a^P b^P, \quad A = \{ab, aabb, aaabbb, \dots\}$$

Assume pumping length = 4

$$\therefore S = aabb$$

Divide S in $x y z$

$$S = \underline{aabb} \quad p=4$$

$\swarrow \quad \downarrow \quad \searrow$
 $x=a \quad y=ab \quad z=b$

$$(1) |xy| \leq p$$

$$|aab| \leq 4$$

$$3 \leq 4 \quad \checkmark$$

$$(2) |y| > 0$$

$$|ab| > 0$$

$$2 > 0 \quad \checkmark$$

$$(3) xy^i z \in A, \quad i \geq 0$$

$$a(ab)^i b \in A \quad ?$$

$$a(ab)^0 b = ab \in A$$

$$a(ab)^1 b = aabb \in A$$

$$a(ab)^2 b = aaabbb \notin A \quad \times$$

$\therefore$ given language $A = \{a^n b^n \mid n \geq 0\}$ is not regular language

Q. Prove that $L = \{0^n 1^n \mid n \geq 0\}$ or $\{a^n b^n \mid n \geq 0\}$ is not regular.

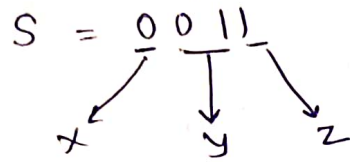
Step 1: Assume that L is regular

Step 2: Consider a string $S = 0^n 1^n \in L$, $|S| \geq p$
where, p = pumping length

i.e. $L = \{\epsilon, 01, 0011, \dots\}$

$S = 0011$, $|S| \geq 4$

By pumping lemma, we can write $S = xyz$



(i) $|xy| \leq p$

$|0011| \leq p$

$|0011| \leq 4$

$3 \leq 4$ ✓

(ii) $|y| > 0$

$|01| > 0$

$2 > 0$ ✓

(iii) $xy^i z \in L, \forall i \geq 0$

let $i = 2$

$0(01)^2 1$

$0(01)^2 1$

$001011 \notin L$

$\therefore L$ is not satisfying pumping lemma

→ This contradicts our assumption that L is regular language.

$\therefore L$ is not regular language.